What Is New In Our Paper?

How our work is different from the research papers we referred to

Explained in simple words

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The Question This Sheet Answers

When you write a research paper, the first thing anyone asks is: **"Other people already worked on this. So what did YOU add?"**

This sheet answers that in plain language. It also tells you honestly which parts are **not** new, so nobody can catch you out.

Part 1: What The Other Papers Already Did

We referred to **7 published research papers**. Here is what each one had already figured out, one line each:

Paper	What they already did
Puzyn, 2011	Proved for the first time that you *can* predict toxicity using a computer instead of a lab
Liu, 2011	Changed the question from "how toxic?" to a simpler **"toxic: yes or no?"**
Oh, 2016	Collected hundreds of studies together and showed the surface and the test conditions matter, not just the material
Trinh, 2018	**Cleaned up** nanoparticle datasets and checked their quality before building models
Furxhi, 2019	**Compared many models** against each other to see which type works best
Furxhi, 2020	Reviewed the whole field and **complained about common mistakes**: tiny datasets, unfair testing, too much focus on accuracy
Subramanian, 2021	Built the **dataset we used**, plus one model of their own

> **Notice the gap:** Trinh cleaned data. Furxhi compared models. But **nobody did both on the same dataset** — and nobody checked what happens when you clean the data *and then* compare the models. That gap is where our work sits.

Part 2: The Three New Things We Did

1 The main one: fixing one problem secretly created another

This is our most interesting finding. Follow it slowly, it has two parts.

Part A — the answer was hidden in the data

The dataset had a column called *viability* (how many cells survived). But the toxic/safe answer was **calculated from that exact column**. So if the computer sees it, it is not predicting — it is just reading the answer.

> **Like this:** giving a student an exam with the answer key accidentally printed on the back page.

We deleted that column. Problem solved — or so we thought.

Part B — deleting it caused a brand new problem

Scientists repeat the same experiment several times. Those repeat rows were **almost** identical — they only differed in the *viability* number.

So the moment we deleted that column, those rows became **exactly identical**. **122 duplicate rows appeared out of nowhere.**

> **Why that is bad:** we split the data into "practice questions" and "exam questions." With duplicates around, **the same question ends up in both**. The computer sees the exam paper during practice. Its score looks great, but it cheated without anyone noticing.

> **The point: fixing problem 1 is what created problem 2.** Most people would fix the first one, feel satisfied, and never look again. That chain reaction is the new thing we are reporting.

And it actually mattered:

	Before we fixed it	After we fixed it
Which model won	Decision Tree	**Random Forest**
Its score (F1)	0.800	**0.865**
Toxic ZnO caught	26%	**72%**

The Decision Tree had only been "winning" because it was quietly memorising the duplicate rows. Fixing the data **changed our final answer.**

2 We tested the model on a material it had never seen

Normally you shuffle all the data and split it randomly. But our 5 materials each have their own fixed chemistry values — so the model can secretly learn **"this is ZnO, and ZnO is usually toxic"** instead of learning real chemistry.

So we ran a harder test: **train on 4 materials, then test on the 5th one it has never seen at all.**

> **Like this:** a student who scores well on the chapters they revised, versus a student given a question from a chapter they have never opened. The second test tells you far more.

> **The result:** accuracy dropped from **94% down to 77%**. The normal random test was hiding this completely.

Furxhi's 2020 review *complained* that people test unfairly. We actually did something about it and **measured exactly how big the gap is.**

3 We compared five models on this particular dataset

The team who built this dataset (Subramanian, 2021) made **one** model. We tried **five** and compared them fairly.

We also judged them the right way. We did **not** pick the winner by accuracy, because with only 1 in 5 particles being toxic, accuracy lies.

> **Proof that accuracy lies:** the KNN model scored **87.6% accuracy** — looks good! But it caught only **41%** of the toxic particles. It missed nearly 6 out of every 10 dangerous ones. If we had judged by accuracy we might have picked it.

Part 3: What Is NOT New — Be Honest About This

If you claim too much, an examiner will catch it. Here is what you should **not** claim:

Do NOT say this	Say this instead
"We invented the idea of data leakage"	It is a well-known concept in machine learning. **Our specific example** is what we found
"We invented this way of testing"	Holding out a whole group is a known technique. **Applying it here and measuring the gap** is ours
"We found a mistake in Subramanian's paper"	They made the label from viability themselves, so they knew. The trap only catches **people who reuse their dataset later**
"Nobody has ever reported this"	"We did not find it in the papers we reviewed." We read 7 papers, not the entire field

> **Why this matters:** being careful about what you claim makes you look **more** confident, not less. Examiners respect a student who knows the limits of their own work. Over-claiming is the fastest way to get picked apart.

Part 4: If They Ask You In The Viva

Q. What is new in your work?

> Earlier papers either cleaned the data **or** compared models. We did both on the same dataset. While doing it we found that removing the leaking column **creates** duplicate rows, and fixing that **changed which model won.**

Q. Why does your duplicate finding matter to anyone else?

> Because this dataset is public and other people will use it. Anyone who removes the leaking column (which they must) will silently end up with 122 duplicates and get a score that is too good. It is a simple check: **look for duplicates again AFTER deleting any column that came from the answer.**

Q. Your accuracy is 94% but also 77%. Which is true?

> Both, for different questions. **94%** is for a material the model has already studied. **77%** is for a completely new material. We report both because quoting only 94% would be misleading.

Q. Why Random Forest and not the others?

> It had the best balance of catching toxic particles without too many false alarms (F1 = 0.865), and it was the most **stable** across repeated tests.

Q. Isn't 355 rows too small?

> Yes, and we say so in the paper. That is exactly why we did the harder material-wise test instead of trusting one lucky split — and why our main suggestion for future work is simply **more materials.**

Remember This One Thing

Earlier papers either **cleaned the data** or **compared the models**. We did both together — and discovered that removing the column which leaks the answer **secretly creates duplicate rows**. Fixing that **changed which model won**. We also showed that on a brand-new material, accuracy falls from **94% to 77%** — a drop the normal testing method hides completely.

That is a genuine, honest contribution for a project of this size. It is not a Nature paper, and it does not pretend to be — but it is real work with a finding that actually changed the result, which many student papers cannot say.